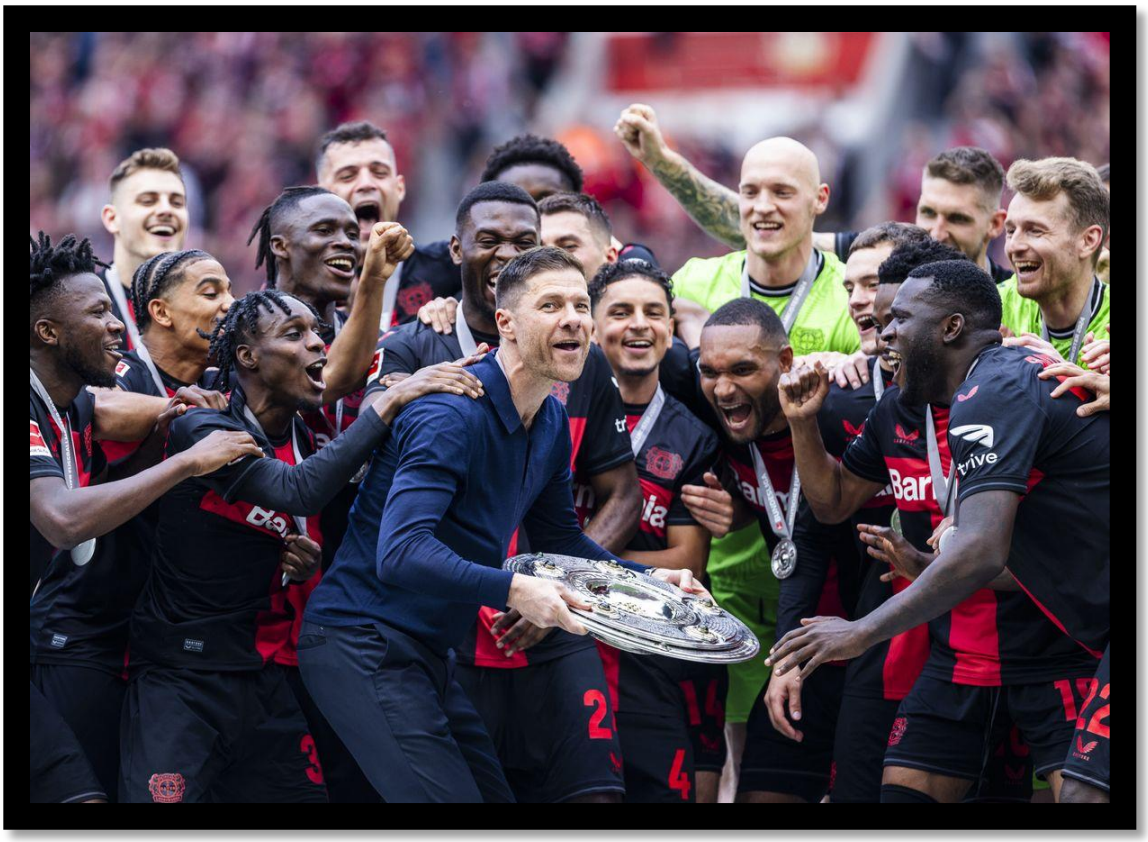

Team Analysis: Bayer Leverkusen 2023/24

What was behind their invincibility?

Author: Zănoagă Cezar-Ionuț

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Introduction

The 2023/24 Bundesliga season was a historic one: Bayer Leverkusen won the league without losing a single match, after Bayern Munich had dominated the competition for the past decade. This achievement raises an essential question:

How did Bayer Leverkusen manage to remain unbeaten throughout the entire season?

The objective of this project is to identify the tactical and statistical factors that contributed to this performance. The analysis focuses on Leverkusen's style of play and compares it with other teams in the league using relevant performance indicators, in order to understand whether their invincibility was the result of offensive dominance, defensive stability, or efficient management of decisive moments.

The working hypothesis is that the team's unbeaten season was driven by a combination of game control, above-average offensive and defensive efficiency, and the ability to manage the final moments of matches.

The analysis is based on StatsBomb event data from the 2023/24 season and evaluates team performance across four dimensions: attacking phase, build-up play, defensive organization, and mentality.

Data Collection and Processing

The source code, datasets, scripts used in the analysis, and the generated visualizations are available in a public GitHub repository:

https://github.com/cezarzng/LeverkusenInvinciblesFootballAnalysis

1. Data Sources

The analysis is based on two main data sources:

- **StatsBomb Open Data (event-level)**
  - Dataset Link: https://www.kaggle.com/datasets/saurabhshahane/statsbomb-football-data
  - Used to describe the team's playing mechanisms, as it contains detailed event data for every action (passes, shots, recoveries, pressures, etc.), including timestamps and pitch coordinates.
  - The data is stored locally in the data/ folder and includes::
    - competitions.json
    - matches/competition_id/season_id.json
    - events/match_id.json

- **Bundesliga 23/24: Team & Player Stats**
  - Dataset Link: https://www.kaggle.com/datasets/whisperingkahuna/bundesliga-2324-team-and-player-insights/data
  - Used for league-level context and comparison.

2. Data Processing

The data processing pipeline is implemented in Python and is divided into four stages:

- **Building the raw dataset**
- **Event extraction & cleaning**
- **KPI calculation / intermediate tables**
- **Visualization generation**

A. Building the Raw Dataset

Events from the StatsBomb JSON files are aggregated into a single raw CSV file

(outputs/datasets/events_raw.csv).

The aggregation script iterates through the matches and concatenates the events from all Bayer Leverkusen matches in the 2023/24 season into a single dataset.

B. **Cleaning and Selecting Event Types**

In the next stage, the events were filtered to a set of types relevant for the analysis (e.g., Pass, Shot, Pressure, Ball Recovery, etc.), while keeping only the columns required for KPI calculation and visualization.

- **Script:** scripts/clean_events.py
- **Output:** outputs/datasets/events_clean.csv

At this stage, three important operations take place:

- Filtering events by their type
- Selecting essential columns
- Normalizing coordinates:
    - location - x, y (start location)
    - pass.end_location - pass_end_x, pass_end_y
    - carry.end_location - carry_end_x, carry_end_y

Finally, the numerical columns (minutes, coordinates, xG, etc.) are explicitly converted to numeric values in order to avoid calculation errors.

C. KPI Calculation and Intermediate Datasets

After the cleaning stage, KPIs are calculated from events_clean.csv using dedicated scripts for each chapter, and the results are exported to outputs/datasets/ as separate CSV files.

For each chapter, the scripts:

- Filter only the team's events
- Calculate the required KPIs
- Export the KPIs to CSV files to be used by the visualization scripts

Four scripts are used to calculate specific KPI sets:

- kpi_offensive.py (Goals vs xG, shot map, set pieces vs open play goals)
- kpi_build_up.py (Progressive actions, progressive actions per player, location data used later for the heatmap)
- kpi_defence.py (Recoveries per player, pressure per player, duels won)
- kpi_mentality.py (Goals by time intervals and points gained after minute 80)

For league-level comparisons, CSV files already available in the **Bundesliga 23/24: Team & Player Stats** dataset were used.

Examples of Calculated KPIs:

- Points gained after minute 800
    - Match information for Bayer Leverkusen was extracted from events data
    - The score at minute 80 and the final score were identified
    - By comparing the two, it was determined whether Leverkusen gained 1 or 3 points after minute 80. If the score remained unchanged, it was considered 0 points gained after minute 80
    - All points gained in this way were then aggregated.

- Finishing efficiency:
    - to calculate this, the number of goals scored was divided by the number of expected goals (xG)

- Progressive actions by location:
    - Pass and carry actions were extracted from the events dataset, including information about the starting and ending locations of each action.
    - Any event that advanced the ball by at least 10 units on the x-axis was considered progressive.
    - The resulting data was exported to a CSV file, which served as the basis for the progressive actions heatmap visualization.

D. **Visualization Generation**

Visualizations were used as the main tool for interpreting the analysis. Their purpose was to transform numerical indicators into clear graphical representations that highlight tactical patterns and differences in an easy-to-understand way.

- Several types of visualizations were used in the project, depending on the nature of the analyzed data:
- League-level comparison charts (bar charts) to position Leverkusen's performance relative to other teams
- Scatter plots to analyze relationships between indicators (e.g., goals conceded vs xGA)
- Heatmaps to represent progression and recovery zones on the pitch
- Shot maps to identify shooting locations
- Player-level charts to highlight players with significant impact in the analyzed phases

The choice of visualization type was made according to the analytical question being addressed. Comparative charts were used to provide league context, while heatmaps helped highlight tactical behaviors.

The visualizations were generated using Python, through the matplotlib and mplsoccer libraries. The choice of these tools allowed a clear representation of both

classic statistical indicators and football-specific analytical elements, such as pitch maps.

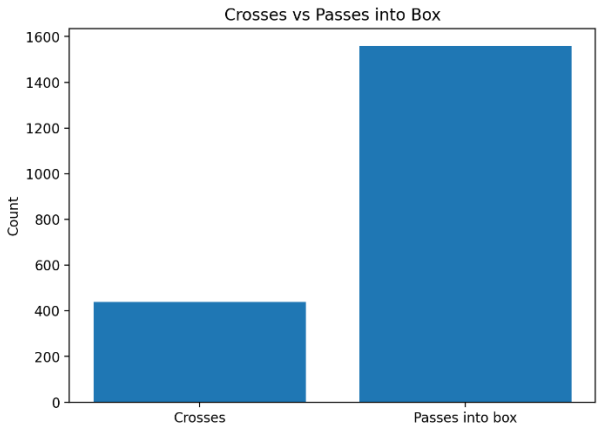
3. Data Validation

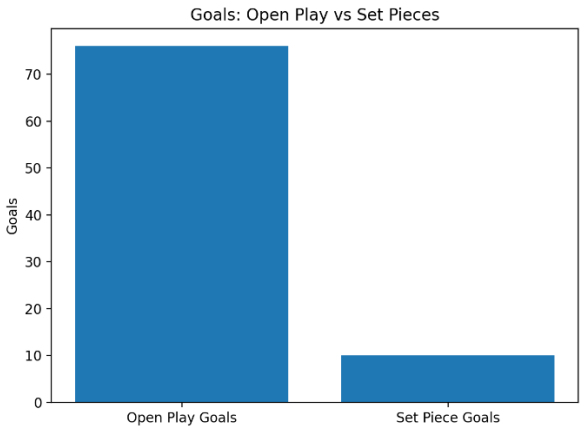
To validate the data and visualizations, the following checks were performed:

- the existence of event files for all **34 matches** played by Bayer Leverkusen
- the presence of all required columns for the defined KPIs
- consistency of team and player names
- absence of missing values (missing values were replaced with values that would not affect the analysis)
- correct representation of the data in the generated charts

Offensive profile

1. Attack Typology





During the 2023/24 season, Bayer Leverkusen's attacking style relied primarily on combinations and central penetrations, rather than repeated crosses from wide areas. The number of passes into the box was significantly higher than the number of crosses, indicating a positional build-up approach focused on central progression.

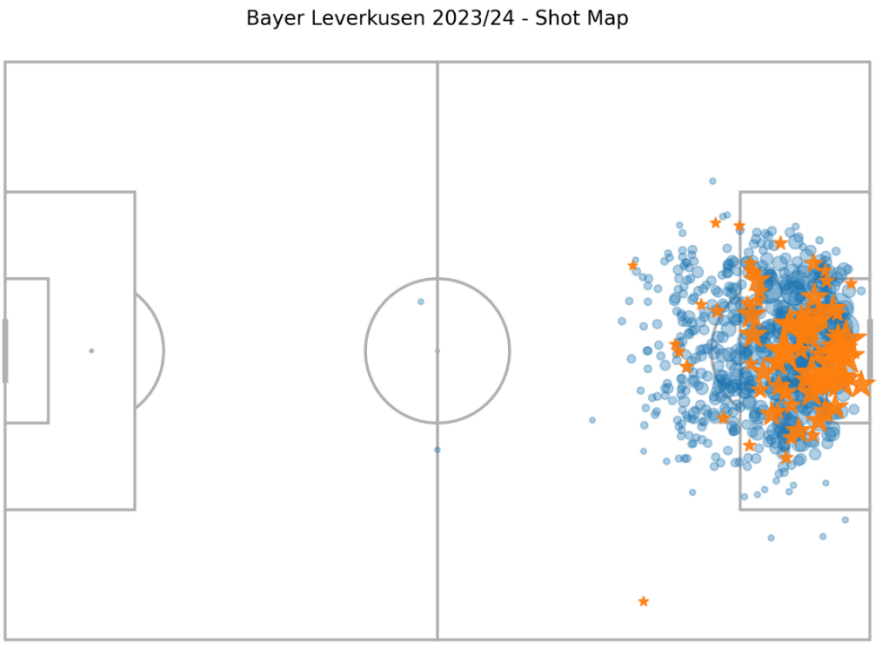
In addition, most of the goals were scored from open play rather than set pieces. This suggests that Leverkusen was able to consistently create and finish chances through structured attacking play, not only through dead-ball situations.

2. Finishing zones

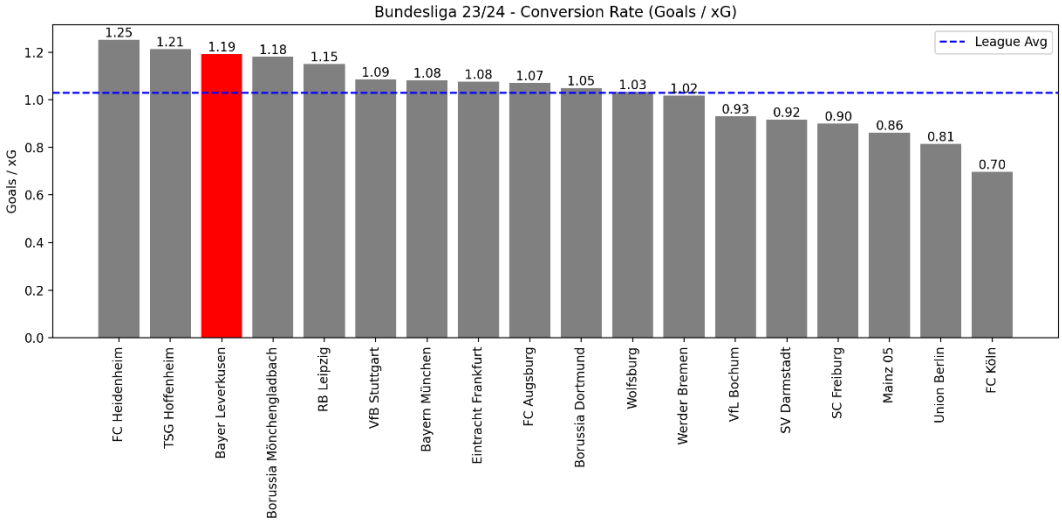
The shot map shows that the majority of attempts came from inside the penalty area, especially from central zones. This indicates that Leverkusen was able to create high-quality chances close to goal.

At the same time, a number of shots were also taken from outside the box, showing that the team had the ability to finish from longer distances as well.

Overall, the shot distribution confirms a well-organized attacking structure, capable of penetrating dangerous areas consistently.



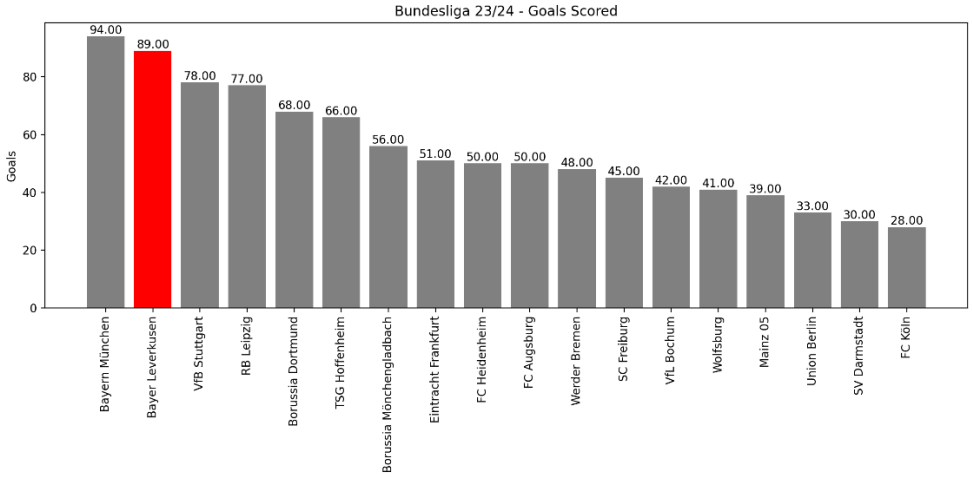
3. Offensive Efficiency



In terms of efficiency, Leverkusen ranked among the most effective teams in the Bundesliga. Their conversion rate (1.19) was above the league average (1.03), indicating that the team converted chances at a higher rate than expected based on xG.

In terms of the **total number of goals**, Leverkusen had the **second-best attack in the league**. This combination of attacking volume and efficiency represented one of the team's main strengths during the 2023/24 season.

Compared to other teams in the league, Leverkusen not only created many chances, but also **converted them efficiently**.



Key Players in Attack:

A remarkable aspect of the Bundesliga season was that the top two assist providers both came from Bayer Leverkusen. Alejandro Grimaldo (13 assists) and Florian Wirtz (11 assists) were among the most important players in chance creation for Leverkusen and also led the league in this category.

Build-up Play

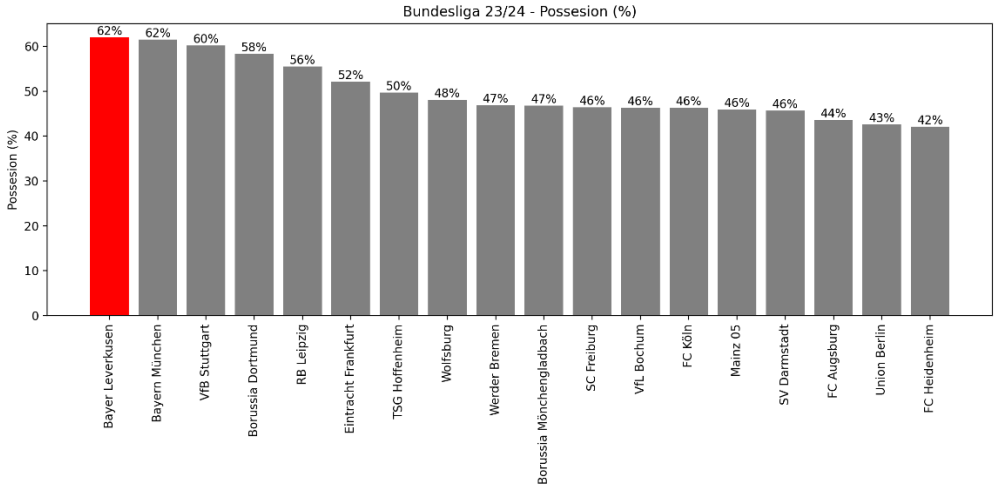
1. Base structure

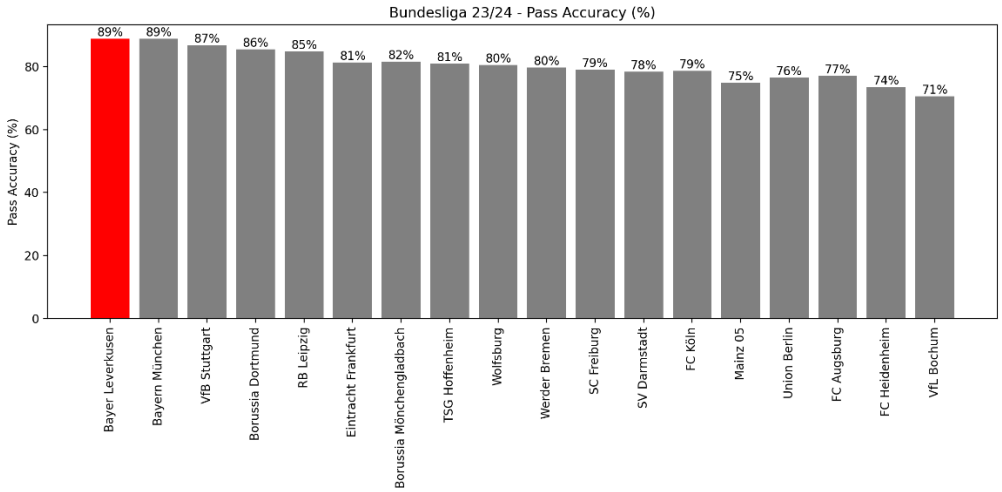
During the season, Bayer Leverkusen operated in a 3-4-2-1 system. The team relied on a back three (Hincapié, Tah, Tapsoba), with two central midfielders in front of them (Xhaka and Andrich).

Grimaldo and Frimpong provided width on the flanks and were actively involved in ball progression, while Wirtz and Hofmann occupied more advanced positions, helping the team move the ball into dangerous areas.

This structure allowed the team to build up play in an organized way, always having multiple passing options across different areas of the pitch.

2. Possession and Passing Accuracy



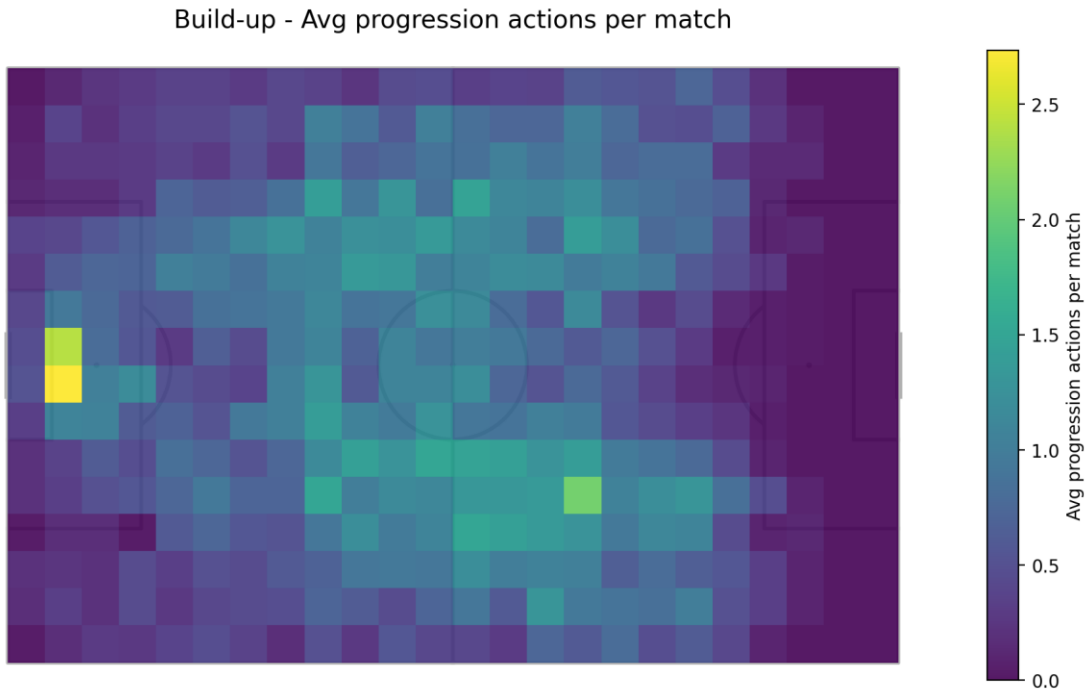


Compared to the other teams in the Bundesliga, Leverkusen had the highest possession rate in the league (62%), tied with Bayern Munich in the 2023/24 season. This allowed the team to control most matches and dictate the tempo of play.

In addition, Leverkusen recorded the highest pass accuracy in the league (89%), also tied with Bayern Munich. This indicator confirms that the team not only circulated the ball frequently, but did so efficiently and with very few errors.

Through these two elements — high possession and passing precision — Leverkusen demonstrated a style of play based on control, organization, and patience in build-up..

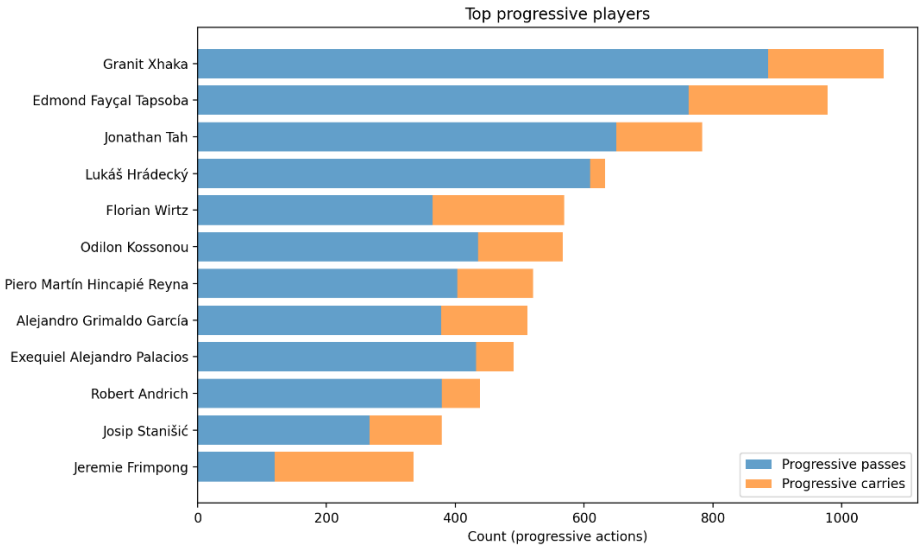
3. Team Progression Zones



he progression zone analysis shows that Leverkusen mainly built attacks through central areas and half-spaces. The high intensity of progression actions in the middle third confirms the team's ability to advance the ball in an organized and controlled manner toward the opponent's goal.

At the same time, the heatmap reveals a balanced left–right distribution, indicating a team that was difficult to anticipate and capable of attacking through multiple zones of the pitch.

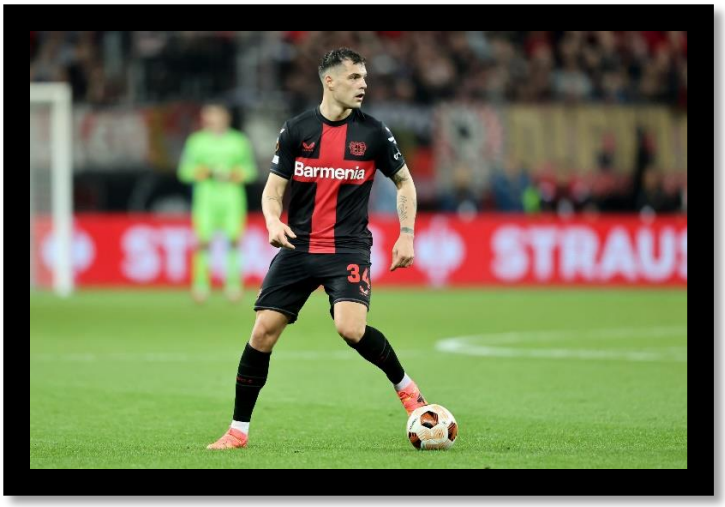
4. Key Players in Build-up



From an individual perspective, Granit Xhaka was the main player responsible for ball progression. He recorded the highest number of progressive passes in the team and represented the central point of the team's build-up structure.

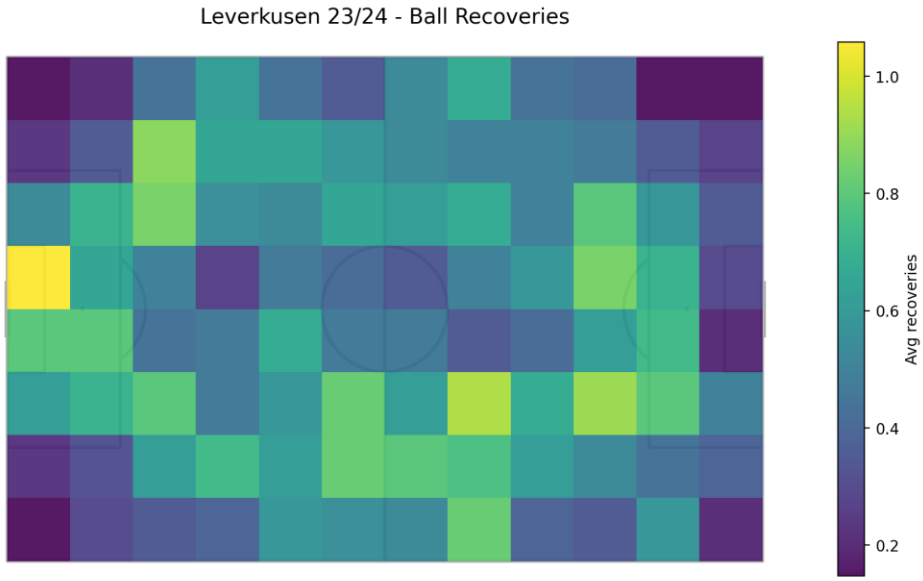
In the defensive line, Jonathan Tah and Edmond Tapsoba played an important role in initiating attacking phases, consistently contributing to progression from the first line.

In the attacking unit, Florian Wirtz was the player linking build-up and finishing, frequently involved in progressive actions.



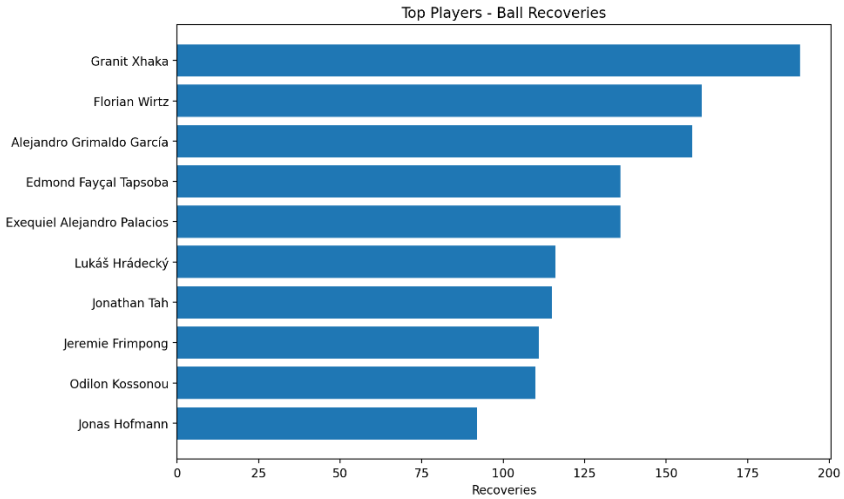
Defensive Profile

1. Ball Recoveries and Intervention Zones



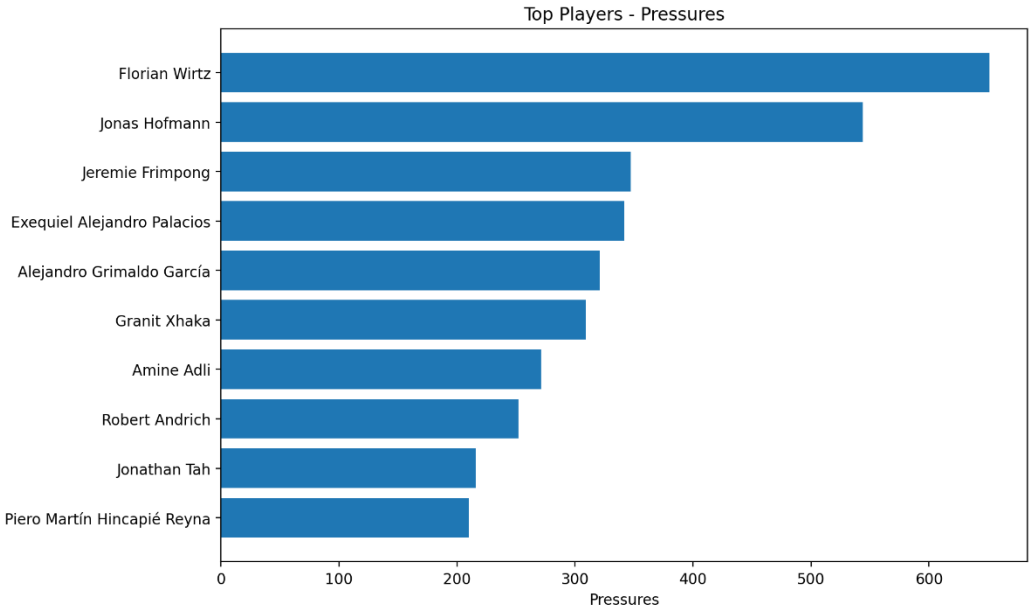
The analysis of ball recoveries shows that Leverkusen had a balanced distribution of interventions across the pitch, with a higher intensity in the middle third and the opponent's half.

The recovery heatmap confirms that the team did not simply wait in a low block, but actively attempted to win the ball higher up the pitch and reduce the opponent's space.



At an individual level, Granit Xhaka, Florian Wirtz, and Alejandro Grimaldo were among the players with the highest number of ball recoveries, indicating a collective involvement in the defensive phase, not only from the back line.

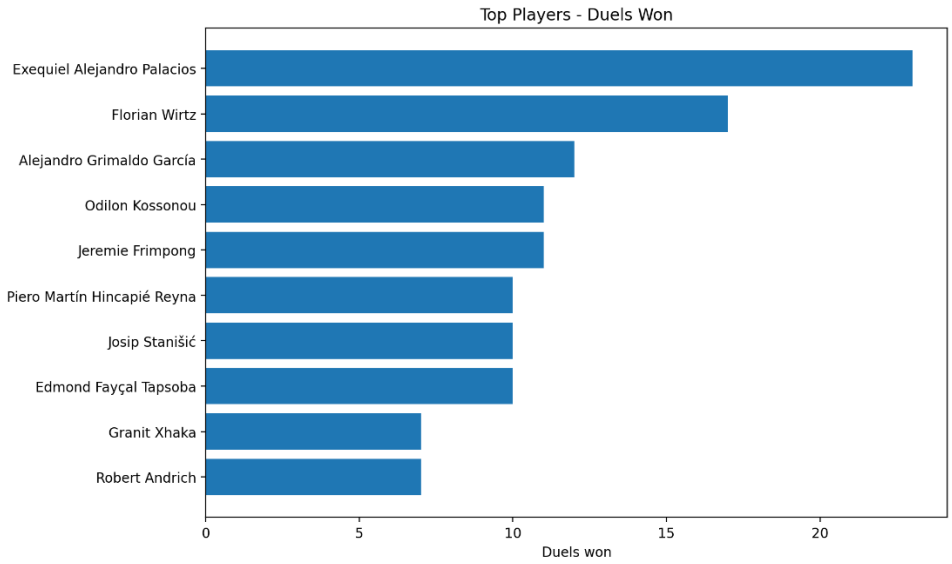
2. Pressing and Duels



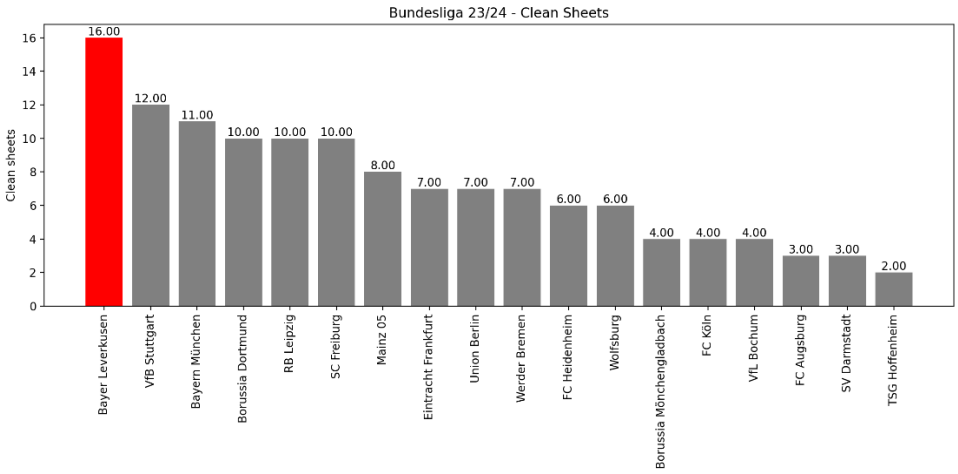
Leverkusen practiced a very active pressing game. Florian Wirtz and Jonas Hofmann stood out for their high number of pressures, showing that the defensive phase often began in advanced attacking areas.

In terms of duels won, Exequiel Palacios, Wirtz, and Grimaldo made important contributions. This highlights that the team was competitive in physical contests and capable of recovering possession quickly.

The combination of pressing intensity and efficiency in duels allowed the team to control the rhythm of matches and limit the opponent's attacking initiatives.

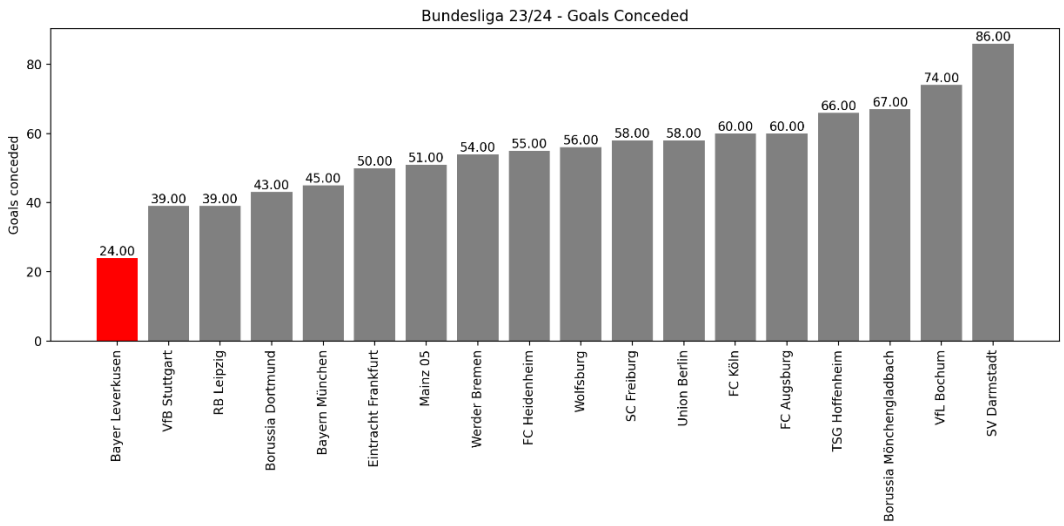


3. Defensive Efficiency



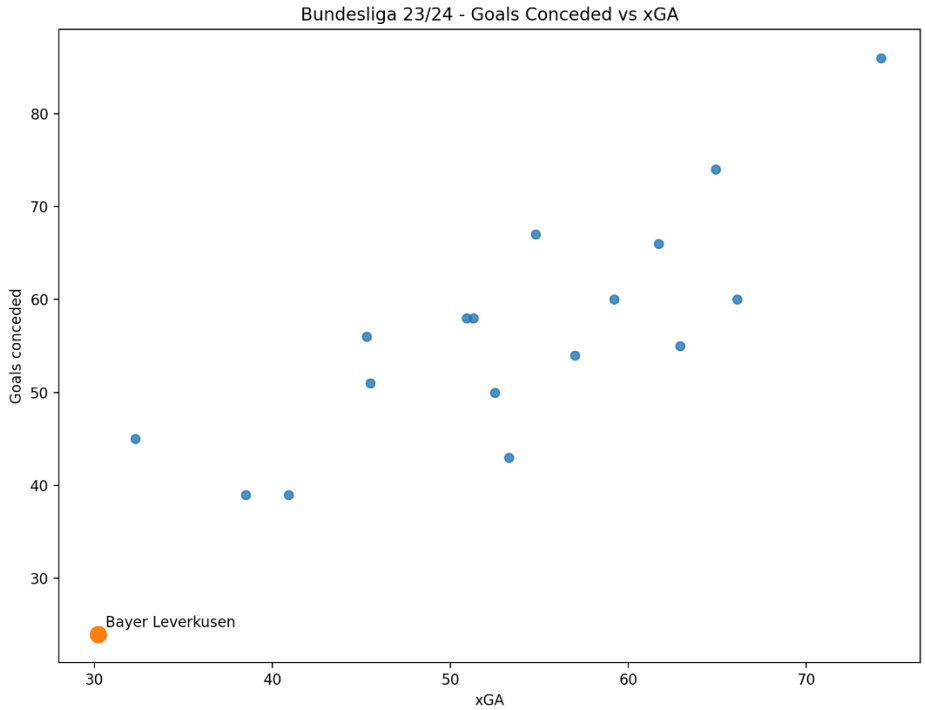
From a statistical perspective, Leverkusen recorded the **highest number of clean sheets in the Bundesliga 2023/24**. This indicator reflects the team's defensive consistency throughout the season.

Additionally, the team conceded the **lowest number of goals in the league**, confirming the overall effectiveness of their defensive system.

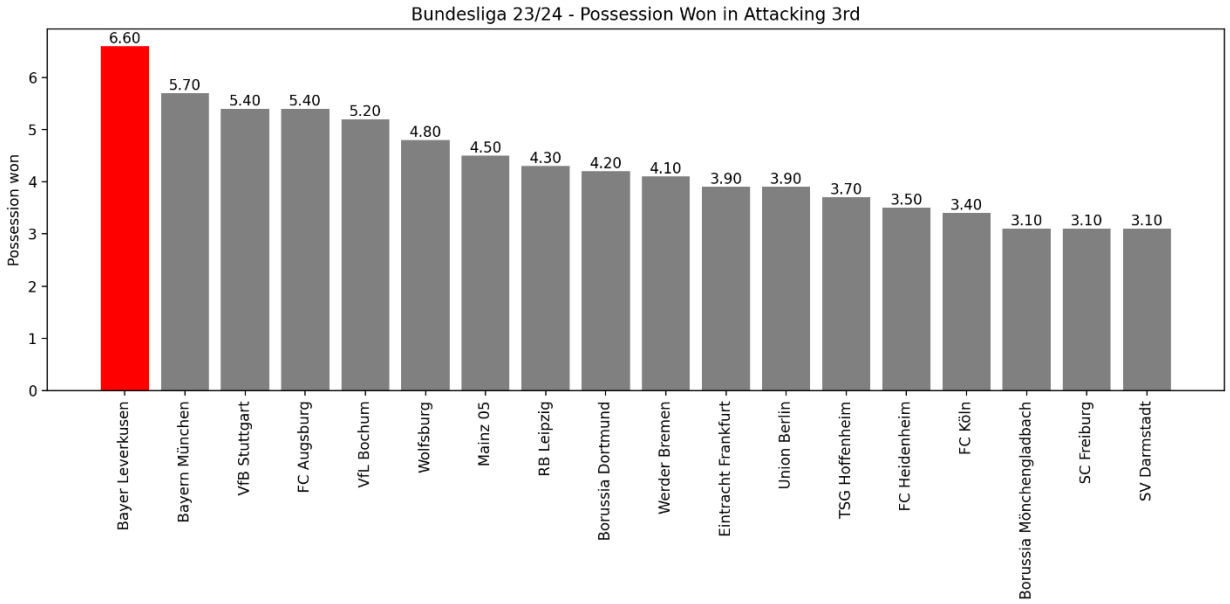


Comparing the goals conceded with xGA (expected goals against), Leverkusen conceded 24 goals, while the expected value was approximately 30, resulting in a difference of –6 goals.

This suggests a high level of defensive efficiency, strongly influenced by both the goalkeeper's performances and the team's collective defensive organization. In this aspect, Leverkusen was clearly the best-performing team in the Bundesliga.



4. High recoveries



A distinctive feature of the team during the 2023/24 season was the high number of ball recoveries in the attacking third. Leverkusen ranked first in the Bundesliga in this metric.

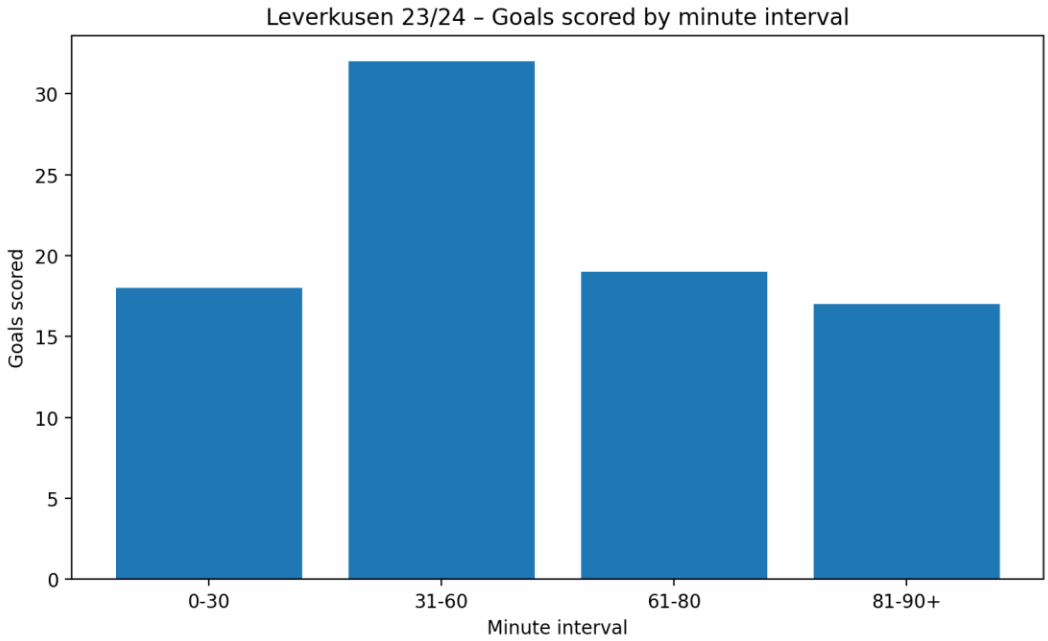
This shows that the team did not only defend effectively in its own half but also attempted to win the ball back closer to the opponent's goal, quickly transforming defensive situations into attacking opportunities.

This aggressiveness contributed to dominating opponents and maintaining constant pressure.

Late Game Performance

A defining feature of Bayer Leverkusen's 2023/24 season was the team's ability to secure points in the final stages of matches.

1. Goals After Minute 80

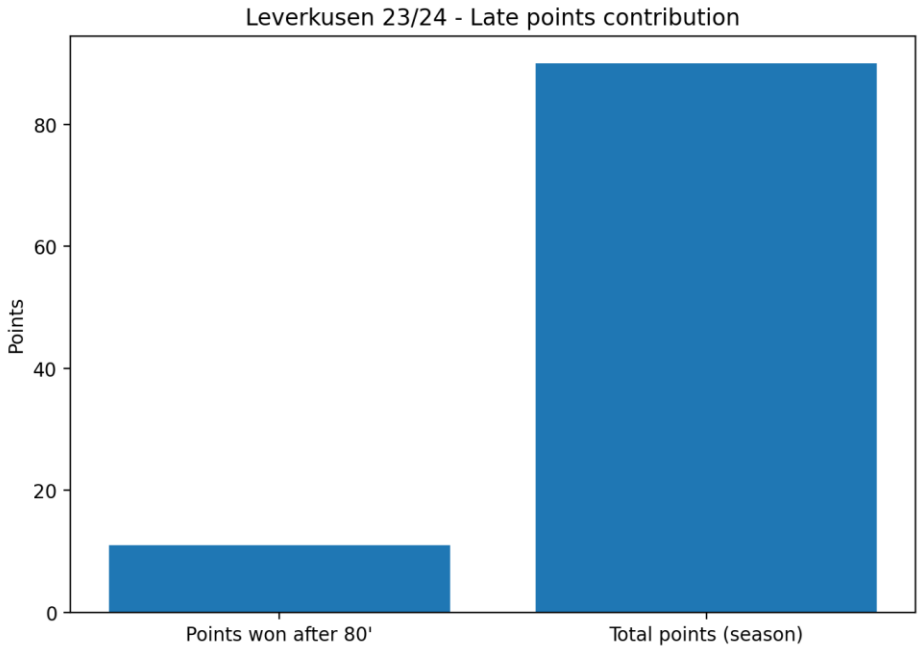


The analysis of goals scored by time intervals shows that Leverkusen remained dangerous throughout the entire match, but continued to score consistently in the final minutes.

The number of goals scored in the 81–90+ minute interval confirms that the team frequently managed matches effectively in their closing stages. This consistency suggests strong physical preparation and high mental resilience.

In the context of the 2023/24 season, these late goals often had a direct impact on match outcomes, contributing to the team earning decisive points.

2. Points Gained After Minute 80



Out of the 90 total points accumulated in the league, a significant portion (11 points) came from goals scored after the 80th minute. This highlights the team's ability to maintain intensity and concentration until the final moments of matches.

The capacity to decide matches late on represented a major competitive advantage and contributed significantly to the team's unbeaten run during the season.

In the 2023/24 season, Bayer Leverkusen not only dominated the Bundesliga statistically but also demonstrated a strong ability to decide matches in crucial moments. The goals scored in the final stages of games played an important role in maintaining an undefeated season.

Conclusion

The analysis of the 2023/24 season shows that Bayer Leverkusen's invincibility was not the result of a single dominant factor, but rather of a structural combination of control, efficiency, and aggressiveness.

From an attacking perspective, the team combined a high volume of actions in central areas with a conversion rate above the league average. The shot distribution and the goals scored from open play confirm a structured attacking model, not dependent on set pieces or isolated situations.

In build-up play, the high possession and passing accuracy indicate systematic control of matches. The 3-4-2-1 structure provided multiple progression options in the middle zones and ensured balanced progression on both flanks.

Defensively, Leverkusen not only conceded the fewest goals in the league, but also outperformed the statistical expectations (xGA), suggesting both strong collective organization and individual defensive efficiency. High recoveries and constant pressure reduced the time opponents spent in possession and favored quick transitions.

Another distinctive feature of the season was the team's ability to decide matches in the final minutes. The 11 points gained through goals scored after minute 80 highlight the team's mental resilience and physical consistency — key factors in maintaining an unbeaten season.

Leverkusen's invincibility was therefore not a statistical accident, but the result of a combination of control, efficiency, mentality, and aggressiveness.

	Club	MJ	V	E	I	GM	GP	DG	Pct.
1	Bayer	34	28	6	0	89	24	65	**90**
2	Stuttgart	34	23	4	7	78	39	39	**73**
3	Bayern	34	23	3	8	94	45	49	**72**
4	RB Leipzig	34	19	8	7	77	39	38	**65**
5	Dortmund	34	18	9	7	68	43	25	**63**